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TECHNICAL CONCEPT NOTE

Hydrogen-Based Proprietary Fuel Retrofit Architecture for Internal Combustion Engine Platforms

Coimbatore, Tamil Nadu, India
August 2026

Proprietary & Confidential, Prepared for Institutional, Regulatory, and Investment Review

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NOTE: This document consolidates HERRSCHER's technology architecture, safety approach, and development roadmap into a single reference for institutional, regulatory, and investment audiences. All technical parameters are indicative at the current pre-prototype / early-prototype stage and are subject to revision based on ongoing laboratory characterisation, simulation, and bench validation. This document does not constitute a type-approval application and does not disclose proprietary formulation or calibration data.

1. Executive Summary

HERRSCHER is an early-stage deep-technology startup, Udyam-registered in Tamil Nadu, India, developing a hydrogen-based, carbon-free fuel retrofit architecture for internal combustion engine (ICE) platforms. The company's objective is to make India's very large installed base of diesel-fuelled stationary gensets and heavy commercial vehicles (HCVs) convertible to near-zero-carbon operation without requiring wholesale replacement of existing mechanical platforms.

At the core of the architecture is a hydrogen-based fuel system built around the HERRSCHER Proprietary Fuel Blend, hydrogen combined with a proprietary, non-carbon combustion-moderating agent developed in-house. The blend is engineered to retain hydrogen's near-zero-carbon combustion product profile while addressing the practical combustion-control challenges, flame speed, thermal loading, and abnormal-combustion risk, that arise when hydrogen is used directly as an ICE fuel. The specific composition and formulation of the blend are proprietary and are treated as a protected trade secret; this document describes the system architecture built around it without disclosing composition-level detail.

The retrofit architecture applies spark-assisted, common-rail direct gaseous injection to a donor compression-ignition short-block, replacing the fuel path, ignition system, and combustion control strategy while preserving the mechanical core of the base engine. This approach is intended to keep retrofit cost, tooling complexity, and platform disruption low relative to a clean-sheet hydrogen engine design.

HERRSCHER's initial research platform is a 25-kW liquid-cooled stationary diesel genset, selected for its representative displacement, controlled operating envelope, and mechanical accessibility for modification. Successful validation on this platform is intended to inform a subsequent extension to heavy commercial vehicle retrofit applications.

Snapshot

Field	Detail
Company	HERRSCHER
Technology Category	Hydrogen-Based Proprietary Fuel Retrofit for ICE Platforms
Initial Platform	25 kW Liquid-Cooled Diesel Genset (Stationary)
Subsequent Platform	Heavy Commercial Vehicle (HCV) Retrofit
Development Stage	Pre-Prototype / Early Prototype, Component and Bench Validation
Fuel Strategy	Hydrogen + HERRSCHER Proprietary Fuel Blend, Spark-Assisted Direct Injection
Base	Coimbatore, Tamil Nadu, India

2. Background and Strategic Rationale

2.1 The Problem with Diesel-Powered ICE Platforms

Diesel-fuelled internal combustion engines remain the dominant prime mover for off-grid stationary power generation and heavy commercial transport across India. Despite improvements in fuel injection, engine management, and after-treatment under progressively tightening CPCB norms, compression-ignition diesel platforms continue to present a consistent set of structural problems:

- Significant CO₂ emissions per unit of energy delivered, inconsistent with India's Paris Agreement NDC commitments and net-zero trajectory.
- Elevated particulate matter (PM_{2.5}, PM₁₀) and NO_x emissions, particularly under transient load and cold-start conditions.
- Dependence on imported crude-oil derivatives, exposing operators to price volatility and supply-chain risk.
- A regulatory trajectory (CPCB-III and successors) that will progressively raise the compliance cost of diesel-based generation and haulage.
- Increasing scrutiny of diesel emissions in urban, hospital, data-centre, and industrial settings.

2.2 Hydrogen as an Energy Carrier

Hydrogen is globally recognised as a strategically important energy carrier for decarbonising sectors that are difficult to electrify directly, and stationary power generation and heavy transport are both identified as priority sectors under India's National Hydrogen Mission (NHM). From a combustion standpoint, hydrogen offers a combination of properties that make it attractive as an ICE fuel while simultaneously introducing distinct engineering challenges that any credible retrofit architecture must address directly rather than assume away.

Property	Typical Value	Engineering Implication
Lower Heating Value	~120 MJ/kg	High specific energy by mass
Volumetric Energy Density (gas)	~10 MJ/Nm ³	Requires high-pressure storage; drives onboard storage and refuelling design
Laminar Flame Speed	~2.0–3.0 m/s	Materially faster than petrol (~0.3–0.4 m/s); combustion phasing and control must be engineered accordingly
Auto-Ignition Temperature	~585°C	Compatible with a spark-assisted ignition strategy
Flammability Limits	4–75% v/v in air	Wide range; requires disciplined leak detection and ventilation design
Combustion Product	Water vapour only	No carbon emitted at point of use

2.3 Why a Blended Approach

Hydrogen's wide flammability range and high flame speed create real challenges for stable, controllable combustion in a direct-injection ICE, particularly under transient load. Rather than treat these as secondary concerns, HERRSCHER's architecture addresses them structurally: the HERRSCHER Proprietary Fuel Blend

combines hydrogen with a proprietary combustion-moderating agent selected to temper flame speed and localised thermal severity while adding negligible carbon to the combustion product stream. The formulation, blend ratio, and selection rationale for this agent are the primary subject of HERRSCHER's ongoing fuel-characterisation research and are held as proprietary.

3. Technology Architecture Overview

The HERRSCHER retrofit architecture is organised into five integrated subsystems. Each subsystem is engineered as a discrete, testable unit, with defined interfaces to its neighbours, so that component-level validation can proceed ahead of full engine-level integration.

3.1 Fuel Storage Subsystem

Onboard storage of the HERRSCHER Proprietary Fuel Blend in a hydrogen-rated, high-pressure gaseous storage vessel. Storage pressure is engineered to maintain adequate margin over in-cylinder pressure at the point of injection across the engine's full load range, with the specific operating pressure being finalised through Phase I/II bench characterisation. All storage and handling components are hydrogen-service rated with multi-layer isolation and continuous pressure and temperature monitoring.

3.2 Fuel Delivery Subsystem

A dedicated high-pressure delivery path comprising isolation valves, regulation, filtration, a flashback-rated fuel rail, and pressure and temperature instrumentation, terminating at the injector interface. The delivery path is designed on a defence-in-depth basis, no single component is intended to represent a sole point of failure between storage and the combustion chamber.

3.3 Combustion System (Modified Engine)

A modified cylinder head and combustion chamber geometry integrating hydrogen-compatible direct-injection injectors and a spark-assisted ignition system, replacing the donor engine's compression-ignition combustion strategy while preserving the mechanical short-block.

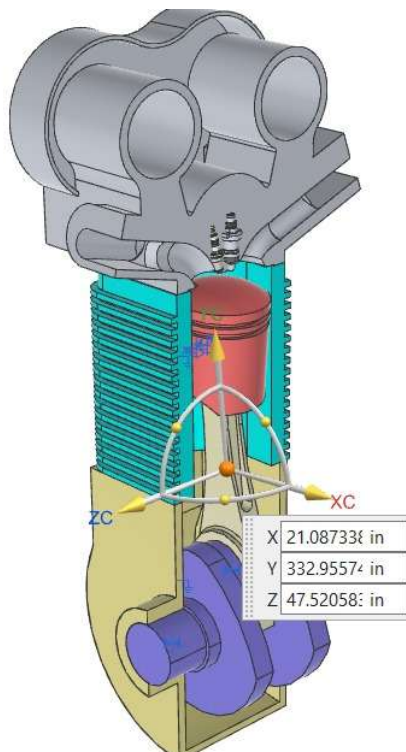


Figure 3.1 , Combustion chamber, piston, and spark plug integration reference (internal CAD model, indicative geometry, subject to revision).

3.4 ECU Control Subsystem

A purpose-configured electronic control unit providing closed-loop fuel injection timing and quantity control, ignition timing management, knock detection and response, fault monitoring, and continuous data logging, described further in Section 5.5.

3.5 Safety Subsystem

A multi-layer protection architecture spanning leak detection, pressure relief, automatic isolation, emergency shutdown, and ventilation management, described in full in Section 6.

3.6 System Integration Logic

Fuel Storage → Fuel Delivery (regulate, filter, meter) → Combustion System (controlled combustion) → ECU Control (monitor, optimise) → Safety Subsystem (protect, isolate, shut off). Every stage reports into the ECU/DAQ layer, and every stage has an independent safety interlock that does not depend on correct operation of the stage before it.

System architecture is described at the subsystem and interface level. Component-level layout drawings, sensor maps, and P&ID-level schematics are maintained separately and are available for technical due diligence.

4. Fuel System and Combustion Cycle

4.1 Four-Stroke Cycle Description

- Intake stroke, air alone enters the combustion chamber via the donor engine's existing intake path; no fuel is present in the intake tract.
- Compression stroke, the air charge is compressed at a compression ratio selected jointly for the blend's combustion characteristics and the ignition strategy, raising chamber pressure and temperature ahead of fuel introduction.
- Injection and ignition, the HERRSCHER Proprietary Fuel Blend is introduced late in the compression stroke through a common-rail direct gaseous injection system, followed by spark ignition from an iridium spark plug, timed off the crank position sensor, to initiate controlled combustion of the injected charge.
- Exhaust stroke, combustion products (predominantly water vapour, with trace species characterised during emissions testing) are expelled through the engine's existing exhaust path.

4.2 Reference Specification

Parameter	Specification
Baseline Platform	Diesel-cycle compression-ignition engine (stationary genset; HCV in subsequent phase)
Fuel	Hydrogen + HERRSCHER Proprietary Fuel Blend
Storage	Hydrogen-rated high-pressure gaseous storage; pressure finalised through bench characterisation
Injection System	Common-rail direct gaseous injection (DI)
Ignition Source	Iridium spark plug, crank-position-sensor timed
Injection Timing	Late compression stroke, phased ahead of ignition
Fuel-Path Integrity	One-way check valve at injector inlet; hydrogen-rated seals throughout
Compression Ratio	Optimised jointly with blend properties and ignition strategy (not a fixed carry-over from the donor diesel platform)

4.3 Ignition System

Ignition is provided by an iridium spark plug, selected for its narrow electrode profile and resistance to erosion under hydrogen's fast, high-temperature flame front. Spark timing is referenced to the crank position sensor and phased relative to the completion of injection, so that the fuel charge is adequately conditioned before ignition without allowing premature dispersion below an ignitable concentration.

4.4 Design Note on Compression Ratio

Compression ratio is treated as an open optimisation variable rather than a fixed carry-over from the donor diesel platform. Because compression ratio, injection timing, and ignition strategy interact directly, HERRSCHER's Phase

II simulation work (Section 9) is scoped explicitly to resolve compression ratio jointly with combustion chamber geometry and blend characteristics, rather than in isolation.

5. Technical Modifications and Engineering Scope

The following modifications are proposed on the donor diesel platform. All specifications are indicative at the current stage and will be refined following combustion simulation, laboratory fuel characterisation, and prototype bench assessment.

5.1 Cylinder Head and Combustion Chamber

The existing diesel cylinder head is assessed and modified or replaced to accommodate direct-injection injector ports and spark plug integration. Combustion bowl geometry is revised to suit the flame propagation characteristics of the HERRSCHER Proprietary Fuel Blend, with head material selection accounting for the altered thermal gradients associated with hydrogen-based combustion.

5.2 Fuel Rail and High-Pressure Supply

A dedicated fuel rail compatible with the proprietary blend is designed and integrated, with material selection, seal specification, and fitting standards determined from the operating pressure envelope established during laboratory testing. Hydrogen-rated materials and hydrogen-compatible elastomers are specified throughout.

5.3 Injector Integration

Hydrogen-compatible direct-injection injectors are integrated into the modified cylinder head, sized to required flow rates and injection-pressure specifications, with response characteristics validated during test-bench evaluation.

5.4 Spark Ignition Integration

A spark-ignition system replaces the compression-ignition mechanism. Spark plug specification, electrode geometry, and ignition energy are selected to achieve reliable ignition across the blend's operating envelope.

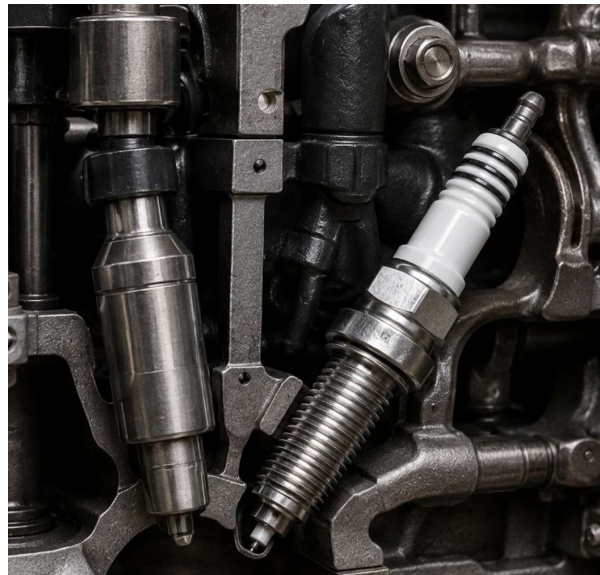


Figure 5.1 , Direct-injection injector and iridium spark plug integration reference.

5.5 ECU and Electronic Control

An ECU providing programmable fuel injection control, spark timing management, closed-loop combustion monitoring, knock detection and response, fault monitoring, and operational data logging is integrated, configured to manage safe startup, shutdown, and emergency-shutdown sequences with diagnostic communication interfaces for data extraction.

5.6 Sensors and Diagnostics

The sensor suite includes in-cylinder pressure transducers, wide-band lambda sensors, intake and exhaust temperature sensors, coolant temperature sensors, fuel pressure sensors at rail and supply lines, knock sensors, hydrogen leak-detection sensors in the engine bay and storage enclosure, and ambient condition sensors, all logged by the ECU and DAQ system.

5.7 Combustion Chamber Optimisation

Compression ratio is optimised jointly with blend properties and ignition strategy, using CFD simulation and empirical test data iteratively to refine chamber geometry, swirl characteristics, and squish architecture.

5.8 Cooling System Enhancements

The base liquid-cooling circuit is assessed and enhanced to manage the altered heat-rejection profile of hydrogen-based combustion, with intake charge temperature management added where test data indicates a requirement.

5.9 Mechanical Component Upgrades

Piston crown geometry, piston ring specification, connecting rod, and bearing specifications are reviewed and upgraded as necessary to withstand the altered mechanical and thermal loading associated with the retrofit, and valve-train components are assessed for hydrogen-service compatibility.

6. Safety Architecture

HERRSCHER's safety architecture is designed on the principles of inherent safety, defence-in-depth, fail-safe operation, and continuous automated monitoring. All provisions described in this section are design objectives subject to experimental validation during prototype development; no statement in this section constitutes a guarantee of absolute safety.

6.1 Governing Hazard Categories

Because HERRSCHER's architecture is a direct-injection system with air-only intake (Section 4.1), the governing hazard categories are those specific to gaseous direct injection rather than the port-injected / premixed failure modes associated with carburetted or manifold-injected hydrogen systems. HERRSCHER's design attention is accordingly concentrated on:

- Surface and pre-ignition risk from hot combustion-chamber surfaces or residual lubricant.
- Injector tip thermal endurance under repeated exposure to hydrogen's fast, high-temperature flame front.
- Seal and seat integrity at the injector and fuel rail under sustained high-pressure gaseous service.
- Reverse-flow / backflow prevention at the injector inlet, addressed structurally through a one-way check valve rather than through injection-timing control alone.
- Leak detection and explosive-atmosphere management across the fuel storage, delivery, and engine-bay zones.

6.2 Leak Detection

Electrochemical and/or catalytic-bead hydrogen leak-detection sensors are installed at all identified potential leak points, including the fuel storage enclosure, fuel rail and high-pressure line junctions, injector mounting interfaces, engine bay interior, and exhaust proximity zones. Alarm thresholds are set at a conservative fraction of hydrogen's lower explosive limit (4% v/v in air), targeting activation at or below 10–20% LEL.

6.3 Pressure Relief and Isolation

Pressure relief valves are incorporated at the fuel storage outlet, fuel rail, and any enclosed high-pressure delivery segment, with discharge routed to a safe external vent location. Electronically actuated, hydrogen-rated isolation valves are installed at the storage outlet and fuel rail inlet, closing automatically on emergency shutdown, ECU fault response, and safe-shutdown sequencing, backed by a manual isolation valve for maintenance and emergency use.

6.4 Automatic Emergency Shutdown (ESD)

An emergency shutdown system, independently wired to the ECU, actuates fuel isolation, ignition cutoff, and engine stop on detection of hydrogen concentration exceeding alarm threshold, fuel pressure anomaly, coolant temperature exceedance, fire or heat detection in the engine bay, or manual ESD activation. The system fails safe in the de-energised state, fuel valves normally closed, ignition off by default on power loss.

6.5 Ventilation Strategy

The engine enclosure and fuel storage area incorporate engineered forced ventilation to prevent accumulation of hydrogen above safe concentration levels, with airflow rates, inlet/outlet positioning, and ESD interlocking designed to hydrogen safety guidelines, using intrinsically safe or explosion-proof rated ventilation equipment.

6.6 ECU-Based Fault Detection and Fail-Safe Modes

The ECU incorporates fault detection covering injector circuit faults, spark circuit faults, sensor out-of-range conditions, communication failures, and hydrogen detection alarms, executing a pre-programmed fail-safe response, up to and including full ESD activation, appropriate to fault severity, with a timestamped fault log for diagnostic review.

6.7 Safe Startup and Shutdown Protocols

Documented startup sequencing includes leak-sensor readiness confirmation, isolation-valve status verification, pressure system pre-check, ventilation confirmation, and ignition readiness check. Shutdown sequencing includes fuel isolation, post-injection purge where applicable, coolant temperature normalisation, and logged power-down.

6.8 Hazard Mitigation Philosophy

The overall safety philosophy follows a hierarchical risk-control approach: inherent safety through design (minimising hydrogen inventory at any point), engineering controls (automated detection, isolation, and shutdown), procedural controls (operator training and documented procedures), and emergency response planning. This hierarchy is formalised through a Hazard Identification (HAZID) and Hazard and Operability (HAZOP) study conducted during prototype development.

6.9 Reference Standards

The safety architecture is designed with reference to applicable Indian and international standards, including but not limited to IS 15793 (safety of hydrogen systems), the IS/IEC 60079 series (hazardous area classification), AIS-039 and AIS-113 (vehicle type-approval provisions relevant to alternative fuels, where applicable), and PESO guidelines for compressed gas storage.

7. Materials Engineering

7.1 Hydrogen Embrittlement Resistance

All hydrogen-wetted components are specified from materials selected for resistance to hydrogen embrittlement. HERRSCHER's material baseline is austenitic stainless steel of the SS316L type for high-pressure hydrogen service, chosen for its low susceptibility to hydrogen-assisted cracking relative to martensitic and ferritic grades, with particular attention to weld zones and components subject to cyclic pressure loading. Lower-pressure runs are specified in appropriately derated carbon steel where hydrogen exposure and pressure conditions permit, consistent with ASME B31.12 materials guidance for hydrogen piping.

7.2 Elastomers and Sealing

Seal and elastomer selection across the fuel path is qualified for hydrogen compatibility and for the thermal and pressure cycling associated with direct-injection operation, with particular qualification attention on the injector seat and high-pressure rail connections.

7.3 Design Assurance Approach

HERRSCHER's materials and safety architecture is developed on the principle that no single component should represent a sole point of failure between the pressurised fuel storage system and the combustion chamber. This includes fuel-path isolation, redundant flow control, and structured component-level testing ahead of engine-level validation.

8. Instrumentation and Data Acquisition

A comprehensive instrumentation and data acquisition (DAQ) system is integrated with the research prototype to support performance evaluation, emission characterisation, combustion analysis, and safety validation.

8.1 Exhaust Emission Measurement

A calibrated exhaust gas analyser measures CO, CO₂, NO_x, unburnt hydrocarbons, and particulate matter at steady-state and transient load conditions.

8.2 Cylinder Pressure Monitoring

High-speed piezoelectric in-cylinder pressure transducers acquire combustion pressure traces, processed to compute indicated mean effective pressure (IMEP), heat release rate, mass fraction burned, and crank-angle-resolved combustion metrics, including cyclic variability (CoV of IMEP) and knock intensity indices.

8.3 Temperature and Fuel Flow Measurement

Thermocouple arrays at intake manifold, exhaust port, exhaust manifold, coolant, lubricant, and cylinder head surface locations characterise thermal distribution, while mass flow meters on the fuel supply line measure instantaneous and cumulative consumption to compute thermal efficiency and specific energy consumption.

8.4 ECU Data Logging and Performance Monitoring

The ECU logs calibration parameters, injection pulse-width, spark timing, sensor inputs, actuator states, and fault codes, extractable via standard diagnostic interface. Engine power, torque, and speed are measured using a calibrated dynamometer or alternator load bank, with generator output voltage, frequency, and power factor recorded for electrical performance assessment.

8.5 Data Traceability

All DAQ channels are synchronised to a common time base and recorded in real time during test sequences, with data files stored in a standard engineering format with traceability to test date, load condition, blend configuration, and operator log.

9. Research and Validation Roadmap

HERRSCHER's development programme runs September 2026 through October 2027, structured around two sequential six-month platform cycles layered on a continuous institutional and growth track. The concept paper, DPR, and funding process run in parallel with prototype build rather than sequentially, so that the programme timeline is not stacked end-to-end.

9.1 Foundation Phase

Core team formation and CSIR-NIIST fuel finalisation anchor Phase 1. In parallel, HERRSCHER develops its concept paper and Detailed Project Report (DPR), publishes a technical paper, and submits a grassroots funding application, targeted to close within three months of programme start.

9.2 Phase 1, Stationary / Genset Retrofit (Sep 2026 – Feb 2027)

A six-month cycle covering prototype build and validation on the 25 kW genset platform, including fuel system adaptation and trials, targeting ARAI certification engagement, with the phase closing by February 2027.

9.3 Phase 2, HCV and Refuelling Systems (Mar 2027 – Sep 2027)

A subsequent six-month cycle extending validated genset learnings to the HCV platform, covering HCV fuel trials, prototype build, and validation / certification, targeting closure by September/October 2027.

9.4 Institutional and Growth Track

Milestone	Timing / Gate
Grassroot funding application	Closes within 3 months of programme start (Sep–Nov 2026)
Phase 1 prototype + validation	Within 6 months of Phase 1 start (Sep 2026 – Feb 2027)
Phase 2 prototype + validation	Within 6 months of Phase 2 start (Mar 2027 – Sep 2027)
IP filing	Filed January 2027; prosecution runs 2–3 years
Team expansion, OEM & H2 partner network	Ongoing through Phase 2
Market entry	Following Phase 2 validation, Q4 2027

9.5 Phased Technical Validation Sequence

- Phase I, Fuel blend characterisation in collaboration with CSIR-NIIST: laminar burning velocity, ignition delay, auto-ignition temperature, quenching distance, and flammability limits as a function of blend ratio.
- Phase II, 3D CFD combustion simulation informed by Phase I data, optimising chamber geometry, injection strategy, and compression ratio; ECU control strategy developed and bench-tested prior to engine integration.
- Phase III, Prototype build and component validation: fuel system pressure integrity, injector characterisation, and ESD functional testing prior to first engine run.

- Phase IV, Engine test-bench evaluation across defined load-speed operating points: power, torque, efficiency, emissions, combustion quality, and safety system behaviour, with durability assessment as data permits.
- Phase V, Safety validation and documentation: fault injection, ESD activation, and leak simulation testing, documented in a Safety Validation Report.

10. Intellectual Property Position

HERRSCHER's competitive position rests on three layers of proprietary work, each developed internally through iterative research rather than adapted from a single published source:

- Proprietary fuel formulation, the composition, blend ratio, and combustion-moderating agent that make up the HERRSCHER Proprietary Fuel Blend, developed and refined through internal fuel-characterisation research.
- System-level integration architecture, the specific combination of injection timing, chamber geometry, quench design, and ignition-control logic calibrated jointly to the proprietary blend's combustion characteristics, rather than to hydrogen alone.
- Execution and calibration know-how, the accumulated bench and prototype calibration data that determines whether a given blend and hardware combination runs stably in practice, which is not reproducible from a formulation alone.

Given the sensitivity of composition-level detail to competitive replication, specific blend ratios, chemical formulation, and calibration parameters are withheld from this document. This is a deliberate protection decision, not an omission, HERRSCHER is actively pursuing formal IP protection on the fuel formulation and select system-integration elements, with filing targeted for January 2027 and prosecution expected to run two to three years. Composition-level and calibration data are available for review under appropriate confidentiality and due-diligence arrangements with prospective partners, incubators, and investors.

11. Future Application Potential

HERRSCHER's current research phase targets a stationary genset platform. The core architecture is designed with scalability in mind, and the following application areas are under active conceptual consideration for subsequent phases.

11.1 Stationary Power Generation

Scaling the retrofit architecture to larger stationary generator sets (>25 kW) serving industrial, commercial, and utility applications is the nearest-term extension, aligned with CPCB emission-reduction targets and the National Hydrogen Mission's stationary power objectives.

11.2 Commercial Vehicle Retrofit

Heavy Commercial Vehicles present a significant decarbonisation opportunity in India. Subject to successful stationary-platform validation, adaptation of the retrofit architecture for HCV engines will be pursued as Phase 2 of the development roadmap (Section 9). Any such application will require separate, full compliance with CMVR, AIS, and ARAI/ICAT homologation requirements before deployment.

11.3 Off-Road and Industrial Equipment

Agricultural, construction, and mining machinery, along with pump sets, compressors, and other mechanical-drive applications powered by diesel ICE platforms, represent further addressable segments for the same modular retrofit architecture, adapted across displacement and configuration.

11.4 Regulatory Prerequisite

HERRSCHER explicitly recognises that all future mobile and on-road applications of this technology are subject to applicable regulatory approvals and certification requirements under the Central Motor Vehicles Rules (CMVR), AIS standards, and ARAI/ICAT type-approval processes. No commercial deployment in mobile applications will be undertaken without prior completion of the required regulatory compliance and certification process.

12. Regulatory Positioning

No standard CPCB or ARAI certification pathway currently exists in India specifically for hydrogen-retrofitted compression-ignition engines. HERRSCHER's approach is accordingly to pursue a validation-dossier-based, case-by-case review pathway, building a structured evidence base from component-level bench testing through to engine-level validation, in consultation with relevant testing and certification authorities as the programme progresses.

12.1 Applicable Standards

HERRSCHER designs to, and tracks evolving guidance under, the applicable Indian and international standards landscape for hydrogen fuel systems, hydrogen-fuelled ICE systems, and compressed-gas storage, including IS, AIS, CMVR, and CPCB frameworks domestically, and ISO, IEC, and SAE frameworks internationally, recognising that the specific regulatory pathway for hydrogen-powered stationary and mobile ICE retrofit in India continues to evolve.

12.2 Engagement Approach

HERRSCHER structures its R&D roadmap to remain aligned with anticipated certification requirements at each phase, engaging test houses, certification bodies, and technical reviewers early in the design process so that engineering decisions made during prototype development do not create avoidable barriers to future certification and commercialisation.

13. About HERRSCHER

HERRSCHER is an early-stage deep-technology startup based in Coimbatore, Tamil Nadu, India, Udyam-registered and founded to accelerate the decarbonisation of India's large installed base of diesel-powered ICE platforms through a certifiable, domestically manufacturable hydrogen retrofit architecture.

13.1 Leadership

- Arun Roshan A S, Founder
- Dinesh Pandi R, Co-Founder

13.2 Mission Alignment

HERRSCHER's technology roadmap is directly aligned with the objectives of India's National Hydrogen Mission, India's NDC commitments under the Paris Agreement, and the broader Green Mobility Initiative , targeting a genuine and pressing national need: decarbonising the large installed base of diesel stationary and mobile ICE platforms without requiring wholesale platform replacement.

13.3 Engagement

HERRSCHER welcomes structured technical review and collaboration from research institutions, testing and certification bodies, OEM and component partners, and investment stakeholders, and is available for a technical presentation and discussion at any interested party's convenience.

14. Frequently Raised Questions

The following questions reflect the kind of technical and commercial scrutiny HERRSCHER's concept has received from reviewing institutions to date, answered here directly so that a reader's core questions are addressed within the document itself rather than deferred.

Why not disclose the exact composition of the proprietary fuel blend?

The blend formulation is HERRSCHER's core protected asset and the basis of its planned IP filing. Composition-level disclosure ahead of filing would materially weaken HERRSCHER's competitive and IP position. Functional behaviour, design rationale, and system-level integration are described in full elsewhere in this document; composition-level and calibration data are available under confidentiality arrangements for qualified technical and investment due diligence.

How is this different from existing hydrogen-ICE and hydrogen-DI work published elsewhere?

Spark-assisted hydrogen direct injection as a general strategy is an active area of published research and OEM development globally. HERRSCHER's differentiation is not the general strategy but the specific proprietary fuel blend and the system-level calibration built around it, the combination of injection timing, chamber geometry, and control logic tuned specifically to that blend's combustion behaviour, which is not reproducible from published hydrogen-only combustion data.

Why start with a stationary genset rather than a vehicle platform?

A stationary platform offers a controlled operating envelope, straightforward test-bench access, and a lower regulatory and safety-validation bar than a mobile application, making it the appropriate first proving ground before HCV-scale validation. This sequencing is reflected directly in the phased roadmap in Section 9.

What is HERRSCHER's plan for addressing the cost gap between hydrogen and incumbent fuels?

HERRSCHER treats fuel economics as a first-order design constraint, not an afterthought. The current phase of work is deliberately scoped to validate technical feasibility on a controlled platform before committing to a specific commercial fuel-sourcing and go-to-market model; site-specific economics, including proximity to existing or byproduct hydrogen sources, are an active part of HERRSCHER's commercial planning alongside the technical validation roadmap.

What happens to the retrofit kit's economics as diesel and hydrogen prices move?

HERRSCHER's retrofit architecture is designed to be fuel-agnostic at the hardware level within the hydrogen-blend family, so that changes in blend ratio or fuel sourcing strategy do not require a redesign of the core injection and ignition hardware, allowing the commercial model to adapt as India's hydrogen supply and pricing landscape matures under the National Hydrogen Mission.

15. Conclusion

HERRSCHER is committed to developing hydrogen-based fuel retrofit technology that is technically sound, practically deployable, and aligned with India's evolving regulatory framework. The company recognises that successful commercialisation of hydrogen ICE retrofit solutions in India requires early, sustained, and collaborative engagement with research institutions, testing and certification authorities, OEM partners, and investment stakeholders.

This document has been prepared as a single, consolidated technical reference, intended to support engagement with government bodies, research institutions, OEM partners, and investors alike, so that the underlying architecture, safety approach, and development roadmap are presented consistently regardless of audience.

The proposed technology addresses a genuine and pressing national need: decarbonising India's large installed base of diesel ICE platforms. HERRSCHER is confident that, with continued technical rigour, appropriate regulatory engagement, and the collaborations described in this document, the proposed research programme can yield validated, deployable technology within the timeline set out in Section 9.

HERRSCHER welcomes technical review of this document and is available for further discussion, a technical presentation, or a due-diligence session at any interested party's convenience.